MILITARY SPECIFICATION

MARKING OF ELECTRICAL WIRES AND CABLES

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the process for marking of identification and designations on electrical wires and cables. Marking is normally accomplished by direct hot stamping on the outer insulation or jacket. Alternate methods involving the use of hot-stamped or ink-stamped identification tapes and sleeves are used in special applications. The hot-stamped markings are inked to provide visual contrast and are impressed for permanency.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

MIL-I-7444 - Insulation Sleeving, Electric, Flexible
MIL-T-713 - Twine, Fibrous, Impregnated, Lacing and Tying

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions, should be obtained from the procuring activity, or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving [his document, should be addressed to: Commander, US Army Armament, Munitions and Chemical Command, ATTN: AMSMC-QAN-Q(D), Dover, New Jersey, 07801-5001, by using the self-addressed Standardization Document Improvement Proposal(DD Form 1426), appearing at the end of this document, or by letter.

AMSC N/A FSC: 1195
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3. REQUIREMENTS

3.1 Designations.

3.1.1 General application. The required designations shall be applied by the method specified on the product drawing using the applicable procedure specified herein.

3.1.2 Manufacturer's designation. When required, manufacturer's code letters will be assigned by the procuring activity and shall be located as shown on the product drawing. In addition, the date code shall be marked by the manufacturer, when required, in accordance with the following system:

"A" for January, "B" for February, etc., "77" for 1977, "78" for 1978. etc. Example: "D78" indicates April 1978 manufacture.

3.2 Hot-stamp marking.

3.2.1 General information. When hot stamping is specified, the outer jacket or sleeving insulation shall be marked using inked marking foils designed for the particular material being marked. The marking shall be impressed to a depth not exceeding one-fourth of the thickness of the material. Markings shall be white on black, dark brown, dark blue, dark red, dark green, and dark purple material. Markings on other colors shall be black. Markings on transparent insulation sleeving shall be black except when the materials underneath are dark colors listed above, in which case the markings shall be white. Unless otherwise specified, markings on white hook-up wires shall be black, Or of a contrast color to white if colored markings are desired to facilitate production. The use of colored markings shall be approved by the procuring activity prior to being used in production.

3.2.2 Equipment and accessories.

3.2.2.1 Marking machine. Hot-stamp marking machines may be either manually operated or automatic and shall be capable of marking the various forms of insulation materials at temperature up to 500° Fahrenheit (°F).

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3.2.2.2 Marking foils. Marking foils must be chosen for compatibility with the materials being marked to effect good adhesion.

3.2.2.3 Type. Type shall consist of vertical capitals of Gothic or similar design. Flat-faced type shall be used for marking on flat surfaces such as on tape and polyvinyl sleeving, and curved-faced type for marking on curved surfaces such as on wires, cables, and sleeving where the type used shall be the largest size which will produce impressed characters on the particular size of material.

3.2.3 Procedure. The marking machine shall be fitted with the correct size type, guide blocks, and foil and then heated to the proper temperature before marking is begun. The section of the insulation to be marked should be well supported and centered under the type and the surface should be as parallel to the face of the type as possible; otherwise, some of the type may cut through the insulation before all the characters are printed. The optimum dwell time and pressure used for each type and thickness of material shall be determined prior to marking the production part. (The operation varialbes must be carefully controlled. A short dwell time of 1 second or less is normally employed except for nylon insulation which requires a dwell time of 1 to 2 seconds. Polyvinyl plastic can be marked at 400° ± 25°F. and polyethylene and synthetic rubber insulation at 375° ± 25°F. Nylon requires a temperature of approximately 450°F. Automatic machines may require higher temperature than hand-operated machines since shorter dwell times are normally employed.)

3.2.3.1 Hook-up wires. Markings on hook-up wires shall be of two types as specified on figure 1. Either type may be employed unless otherwise specified on the product drawing.

Type I - Markings shall be applied recurrently along the length of the wire with at least one marking appearing near each end.

Type II - Markings shall be applied twice at each end of the wire as illustrated. Markings may in addition appear at intervals along the length of the wire.

Markings need not be applied a specific distance from the end of the wire but shall be applied in such a location that the designation is readily visible near the end of the wire after

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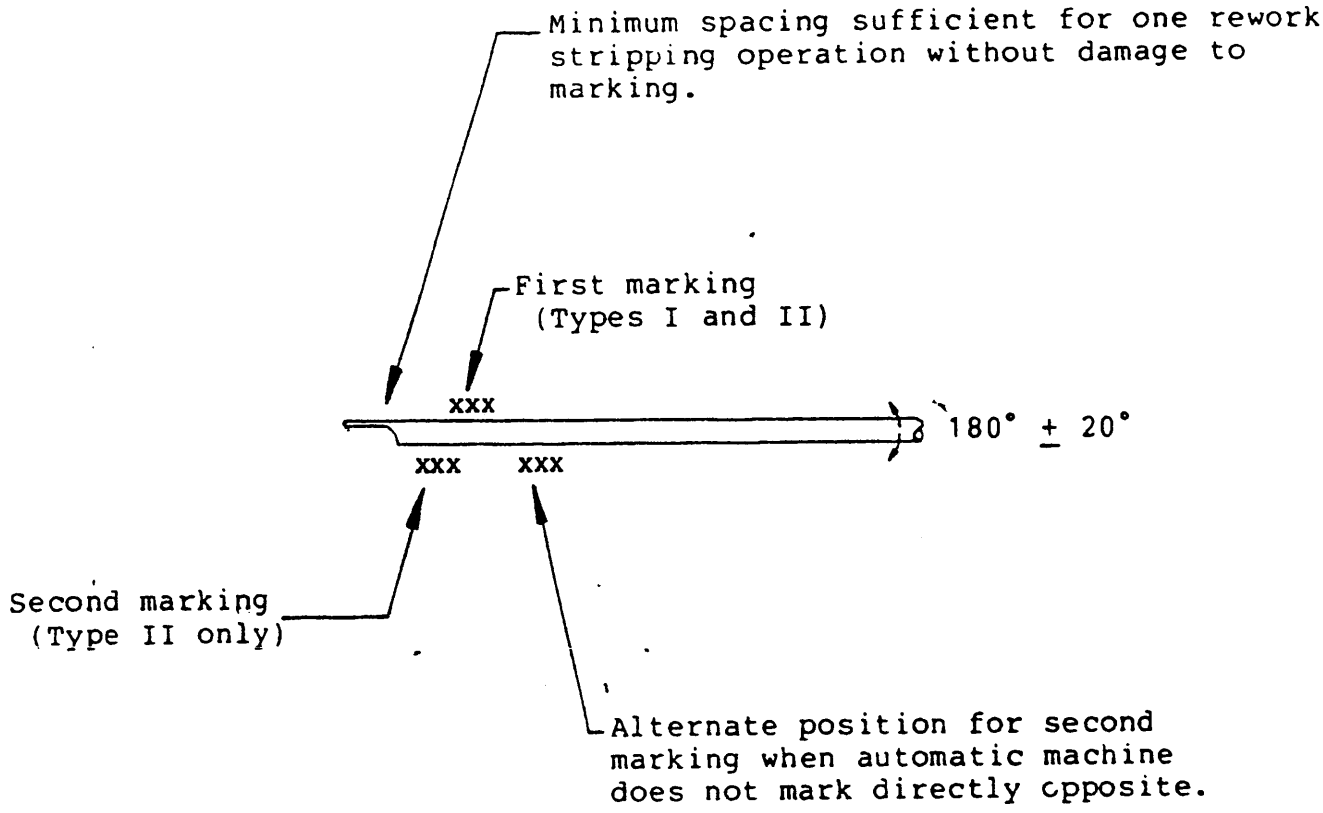


FIGURE 1

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installation. Wires shorter than three inches, such as jumper wires, need not be marked unless specifically designated on the product drawing. If markings on individual wires in harnesses and cables are not required by the product drawing, and such marking would be desirable to facilitate production, manufacturing may hot-stamp suitable designations on the wires.

3.2.3.2 Cables. - Identification designations on cables shall be located on the product drawing and shall consist of a double marking of the designations as shown on figure 2. Coaxial cables and other manufacturer cables with tight-fitting jackets shall be hot-stamped directly on the jacket (see 3.2.3.2.1). When insulation sleeving is employed as an outer jacket over a bundle of wires, the markings shall be applied to the flat sleeving before insertion of the wires.

3.2.3.2.1 Identification sleeves. - Marking of cables with identification sleeves (see figure 3) is a special method which may, when so specified, be used as an alternate to direct hot-stamping of the outer jacket if the following conditions exist: (1) on small cables where experience has shown that the thin insulation is pierced in the stamping operation, (2) on manufactured, multiconductor cables with extruded insulation where, due to the twist and insufficient number of conductors underneath, ridges appear to such an extent that a satisfactory marking cannot be obtained, and (3) on insulation where properties of the material are such that a satisfactory direct marking cannot be obtained. Identification Sleeves shall not be used if a satisfactory marking can be made by direct hot stamping of the outer jacket. Identification sleeves, when used, shall be made from a suitable length of polyvinyl sleeving complying with the requirements of MIL-I-7444. The inside diameter of the sleeve shall be such as to provide a close fit over the cable jacket. The required designations shall be hot-stamped on the sleeve as indicated in 3.2.3.2 prior to placing the sleeve over the cable. On short cable (eg. feet or less), the identification sleeve may extend the full length of the cable.

3.3 Marking with identification tapes. - Printed identification tapes shall comply with the requirements of the appropriate specification referenced on the product drawing. Tapes shall be applied directly around the bundle of wires in the indicated locations prior to inserting the bundle into the transparent insulation sleeving. The tape shall be wrapped one or two turns around one conductor then one and one-half to two turns around the bundle of wires. The required designations shall be ink-stamped or hot-stamped recurrently on the tape at essentially uniform intervals so that the designation will appear at several places around the cable. On small cables the minimum number of markings visible on the installed tape shall be two, located diametrically opposite. The tape shall be wrapped without spiraling and the designations shall be entirely legible through the sleeving.

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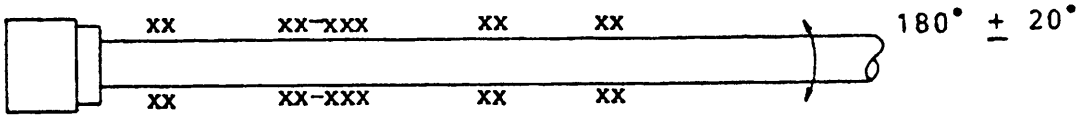
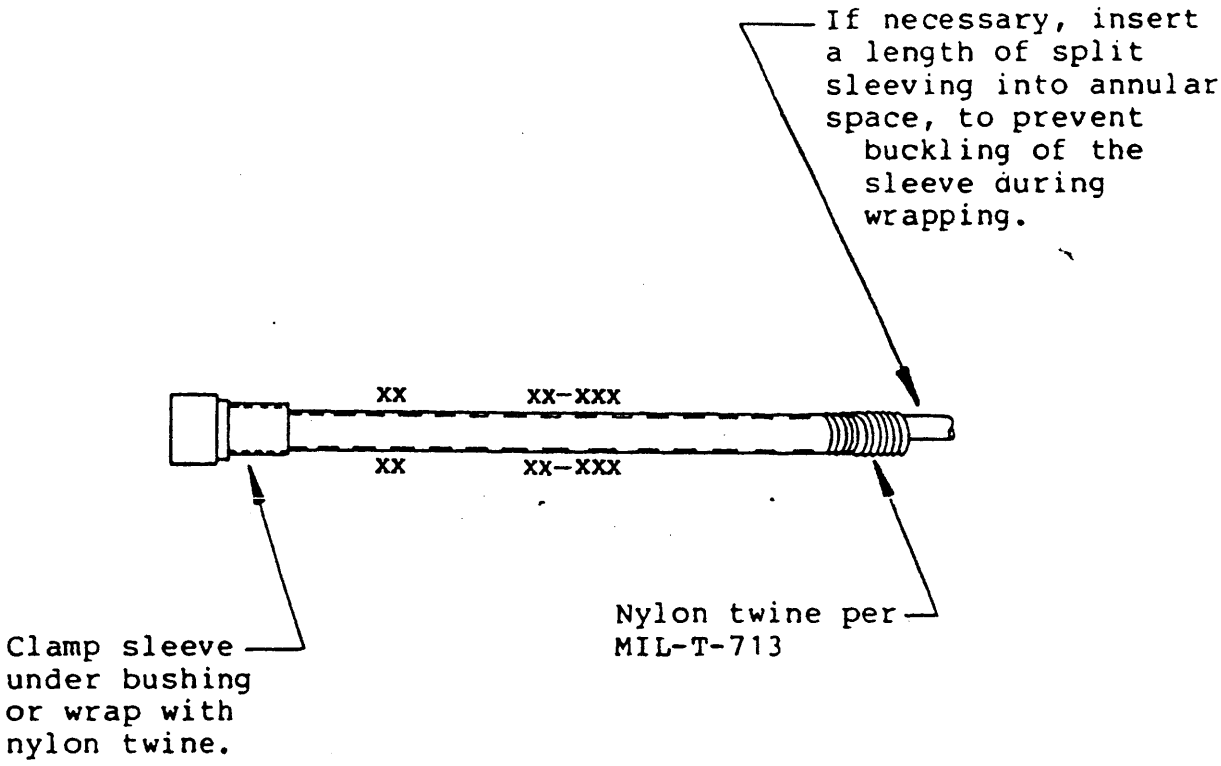


ILLUSTRATION OF MARKING ON CABLES

FIGURE 2

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INSTALLATION OF IDENTIFICATION SLEEVE

FIGURE 3

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3.4 Reworking of markings.

3.4.1 Coaxial and multiconductor cables. Markings that have been hot-stamped directly on the outer jacket of a cable or on an identification sleeve may be reworked by hot-stamping the new "dash" number (eg. -01) at the side of the obsolete designation. The old designation shall not be removed. The surface of the insulation shall be clean before hot-stamping the new designation. Example: obsolete designation, “123456-00”, reworked designation, “123456-00-01.”

3.4.2 Cables with insulation sleeving. In the rework of cables, where it is not necessary to remove the connector or insulation sleeving, rework markings shall be made with identification tapes (see 3.3). The operation shall be performed by loosening the clamp and slipping the sleeving back along the bundle of wires. If the obsolete marking is on tape, the old tape shall be removed and replaced with new tape containing the new designation. If the obsolete marking is hot-stamped on the sleeving, tape printed recurrently with the new “dash” number (eg. -01) shall be applied around the wires in proper location to appear beside and under the hot-stamped marking (eg. xxxxxx-xx-01). The hot-stamped marking shall not be removed. If it is necessary to remove the connector or sleeving during rework. the rework markings may be made with identification tapes, by hot-stamping new designations on the old sleeving in a manner similar to that specified in 3.4.1, or by hot-stamping a new sleeving.

3.5 Workmanship. Hot-stamped markings shall be sharp and clearly defined. The markings shall be applied in such a manner as to be impressed no more than one-fourth the thickness of the material and without damage to the insulation. The ink in the markings shall be adherent. Identification tapes and sleeves shall be neatly applied using good workmanship.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection. The marked material shall be checked to ascertain that the markings are impressed to provide a permanent marking and that the insulation has not been damaged in the marking process. The adhesion of the ink shall be tested by rubbing the

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thumb over the marking. Slight transfer of ink is allowable; however, the general appearance of the marking shall be essentially unchanged by this treatment. The marking shall remain visible by virtue of its impression should the ink be removed from the marking.

5. PREPARATION FOR DELIVERY

5.1 Not applicable.

Custodian
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Preparing activity
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Project No. 1195-A116

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-60903A	MARKING OF ELECTRICAL WIRES AND CABLES

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTICE OF INACTIVATION
FOR NEW DESIGN

INCH-POUND

MIL-M-60903A (AR)
NOTICE 3
21 June 1995

MILITARY SPECIFICATION

MARKING OF ELECTRICAL WIRES AND CABLES

This notice should be filed in front of MIL-M-60903A (AR), dated 14 May 1986

MK-M-60903A (AR) dated 14 May 1986, with Validation Notice 2, dated 20 August 1991 is inactive for new design and is no longer used, except for replacement purposes.

Preparing Activity
Army-AR

AMSC N/A

FSC 1195

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